Notes

1 The Merging Operator

Session Types use a variety of different merging operators depending on their needs. A more expressive merging operator often comes with time complexity tradeoffs. We first disregard this tradeoff, outlining a set of *good* merging operators and show how they satisfy useful properties when tied to projections.

Definition 1.1 (The Merging Operator) *The merging operator $\sqcap$ is a partial binary function over local trees. We write $T_1 \sqcap T_2 = T_3$, which can be read as "T_1 and T_2 merge into T_3". In addition, the merging operator is commutative and associative. Specifically,*

- $T_1 \sqcap T_2 = T_2 \sqcap T_1$
- $(T_1 \sqcap T_2) \sqcap T_3 = T_1 \sqcap (T_2 \sqcap T_3)$

We also assume some more properties that we expect from any merging operator,

- *(Idempotency): $T \sqcap T = T$*
- *(Application): $T_1 \sqcap T_2 = T$, $T_3 \sqcap (T_1 \sqcap T_2) = T_3 \sqcap T$.*

Notation 1.2

We write $\bigsqcap_{i \in I} T_i$ to denote the partial n-ary merging function that merges all the elements in I.

Definition 1.3 (Mergability) *Given some merging function $\sqcap$, if $T \sqcap T' = T$, then we write $T \sqsubseteq T'$ and read "T' is **mergable** into T".*

Definition 1.4 *We call a mergnig operator **good**, if it satisfies the following properties.*

- $(G1)$: *The output of a merging operator maintains structure. That is, if $T_1 \sqcap T_2 = T$, the top layer constructors of T_1 and T are the same.*
- $(G2)$: *The continuations merges as well. That is, if $p?\{\ell_i(S_i).T_i\}_{i \in I} \sqcap p?\{\ell'_j(S'_j).T'_j\}_{j \in J} = p?\{\ell''_k(S''_k).T''_k\}_{k \in K}$ and there exists i, j, k such that $\ell = \ell_i = \ell'_j = \ell''_k$, then $T_i \sqcap T'_j = T''_k$. Similarly for $p!$.*
- $(G3)$: *The relation $\sqsubseteq$ is maintained under constructor composition.*

Lemma 1.5 *If $T_i \sqsubseteq T'_i$ and $\bigsqcap_{i \in I} T_i$ is defined, then $\bigsqcap_{i \in I} T'_i$ is defined and $\bigsqcap_{i \in I} T_i \sqsubseteq \bigsqcap_{i \in I} T'_i$.*

Proof. We note that

$$\bigsqcap_{i \in I} T_i = \bigsqcap_{i \in I} (T_i \sqcap T'_i) = \bigsqcap_{i \in I} T_i \sqcap \bigsqcap_{i \in I} T'_i$$

and as the left side is defined, so is the right side. ■

We want the merging operator to be good in order to prove subject reduction, thus it is also sufficient if we can find conditions for which the properties are satisfied. Note that some literature provide examples of merging operators that don't satisfy $(G2)$ and $(G3)$, including "A Very Gentle Introduction to Multiparty Session Types". However, $(G2)$ and $(G3)$ provides compositionality to the merging operator which behaves well with our proof structure (induction on a prefix of the tree), so we utilize this here.

Proposition 1.6 *Given any unary (linear) context $C[X]$, if for any trees T, T', $T \sqsubseteq T'$ implies $C[T] \sqsubseteq C[T']$, then we satisfy (G3). In fact this is an if and only if condition.*

Proof. This is simply a rephrasing of the definition. ■

2 The Subject Reduction Proof

We assume for the rest of this section that we are working with a good merging operator and that global trees are balanced.

Definition 2.1 *Given a context hole tree $\mathcal{G}$ and an accompanying set of G_i over some indexing set L with $G = \mathcal{G}[G_i]_{i\in L}$, we say that this tree is **compatible** with $p \to q$ if*

- $p, q \notin \mathcal{G}$
- *for all $i \in L$, $G_i = p \to q : \{\ell_i(S_i).G'_i\}_{i\in I}$ for some indexing set I.*

Theorem 2.2 *Given a well formed global type G, if $G' = G\backslash p \xrightarrow{\ell} q$ is defined, there exists a context hole tree $\mathcal{G}$ equipped with the set G_i over an indexing set L with $G = \mathcal{G}[G_i]_{i\in L}$ such that this tree is compatible with $p \to q$.*

Proof. Follows from the fact G is balanced, and that the first instance of p being involved in an interaction is of the form $p \to q$.

Lemma 2.3 *Given a well formed global type G, if $G' = G\backslash p \xrightarrow{\ell} q$ is defined and $G \upharpoonright_p q!\{\ell_i(S_i.)T_i\}_{i\in I}$, $G \upharpoonright_q p?\{\ell'_j(S'_j).T'_j\}_{j\in J}$, then there exists $i \in I$ and $j \in J$ such that $\ell = \ell_i = \ell'_j$ and $G' \upharpoonright_p = T_i$, $G' \upharpoonright_q = T'_j$.*

Proof. Given a well formed global type G, by Theorem 2.2 we can obtain a context hole tree $\mathcal{G}$ equipped with the set G_i over some indexing set L such that

$$G = \mathcal{G}[G_i]_{i\in L}$$

which is compatible with $p \to q$. We proceed our proof by an induction on $\mathcal{G}$.

In the base case, we have $\mathcal{G} = [\ ]_k$ for some k. It follows that

$$G = G_k = p \to q : \{\ell''_k(S''_k).G''_k\}_{k\in K}$$

for some indexing set K. This gives, $G' = G''_k$ with $\ell''_k = \ell$. Then, we know

$$(p \to q : \{\ell''_k(S''_k).G''_k\}_{k\in K}) \upharpoonright_p q!\{\ell_i(S_i).T_i\}_{i\in I}$$
$$(p \to q : \{\ell''_k(S''_k).G''_k\}_{k\in K}) \upharpoonright_q p?\{\ell'_i(S'_i).T'_i\}_{j\in J}$$

It follows that $G''_k \upharpoonright_p T_i$ and $G''_k \upharpoonright_q T'_j$ for $i \in I$ and $j \in J$ such that $\ell = \ell_i = \ell'_j$.

In the inductive case, we have $\mathcal{G} = s \to s' : \{\ell''_k(S''_k).\mathcal{G}_k\}_{k\in K}$. It follows that $G = s \to s' : \{\ell''_k(S''_k).G''_k\}_{k\in K}$ for some G''_k with $G''_k = \mathcal{G}_k[G_i]_{i\in L}$. By compatibility conditions, we have $p, q \notin \{s, s'\}$, so we have

$$G''_k \upharpoonright_p T^{(k)} \quad \text{with} \prod_{k\in K} T^{(k)} = q!\{\ell_i(S_i).T_i\}_{i\in I}$$
$$G''_k \upharpoonright_q T'^{(k)} \quad \text{with} \prod_{k\in K} T'^{(k)} = p?\{\ell_j(S'_j).T'_j\}_{j\in J}$$

Furthermore, $G' = G\backslash p \xrightarrow{\ell} q = s \to s' : \{\ell''_k(S''_k).(G''_k\backslash p \xrightarrow{\ell} q)\}_{k\in K}$. We now focus on the projection with p. As the merging operator is good, by condition $(G1)$,

$$T^{(k)} = q!\{\ell_i^{(k)}(S_i^{(k)}).T_i^{(k)}\}_{i\in I_k}$$

For each $k \in K$, there exists an $i_k \in I_k$ such that $\ell_{i_k}^{(k)} = \ell$, and by condition $(G2)$,

$$\prod_{k\in K} T_{i_k}^{(k)} = T_i$$

where $\ell_i = \ell$. We now note that by the inductive hypothesis, for each $k \in K$,

$$(G''_k\backslash p \xrightarrow{\ell} q) \upharpoonright_p T_{i_k}^{(k)}$$

As $G' = s \to s' : \{\ell''_k(S''_k).(G''_k\backslash p \xrightarrow{\ell} q)\}_{k\in K}$, it follows that

$$s \to s' : \{\ell''_k(S''_k).(G''_k\backslash p \xrightarrow{\ell} q)\}_{k\in K} \upharpoonright_p T_i$$

We can apply the same proof to the case for participant q, and the rest of the proof follows. ■

Lemma 2.4 *Given a well formed global type G, if $G' = G\backslash p \xrightarrow{\ell} q$ is defined and $G \upharpoonright_r T$ for some $r \notin \{p, q\}$, then $G' \upharpoonright_r T'$ with $T \sqsubseteq T'$.*

Proof. Given a well formed global type G, by Theorem 2.2 we can obtain a context hole tree $\mathcal{G}$ equipped with the set G_i over some indexing set L such that

$$G = \mathcal{G}[G_i]_{i\in L}$$

which is compatible with $p \to q$. We proceed our proof by an induction on $\mathcal{G}$.

In the base case, we have $\mathcal{G} = [\ ]_k$ for some k. It follows that

$$G = G_k = p \to q : \{\ell''_k(S''_k).G''_k\}_{k\in K}$$

for some indexing set K. This gives, $G' = G''_k$ with $\ell''_k = \ell$. It then follows that

$$G \upharpoonright_r \prod_{k\in K} T_k$$

where $G''_k \upharpoonright_r T_k$ for all $k \in K$. By the self identity on merging operators, we have $T_k \sqcap T_k = T_k$, which we can extend by application to

$$T_k \sqcap (\prod_{k\in K} T_k) = (\prod_{k\in K} T_k) = T$$

Specifically, we have $T \sqsubseteq T_k$.

In the inductive case, we have $\mathcal{G} = s \to s' : \{\ell''_k(S''_k).\mathcal{G}_k\}_{k\in K}$. It follows that $G = s \to s' : \{\ell''_k(S''_k).G''_k\}_{k\in K}$ for some G''_k with $G''_k = \mathcal{G}_k[G_i]_{i\in L}$. We have three cases to consider:

- $r = s$

We have,

$$r \to s' : \{\ell''_k(S''_k).G''_k\}_{k\in K} \upharpoonright_r s'!\{l''_k(S''_k).T''_k\}_{k\in K}$$

with $G''_k \upharpoonright_r T''_k$ for $k \in K$. By the inductive hypothesis, we know

$$(G''_k \backslash p \xrightarrow{\ell} q) \upharpoonright_r \bar{T''_k}$$

with $T''_k \sqsubseteq \bar{T''_k}$
Noting $G' = r \to s' : \{\ell''_k(S''_k).(G''_k \backslash p \xrightarrow{\ell} q)\}_{k\in K}$, we have

$$r \to s' : \{\ell''_k(S''_k).(G''_k \backslash p \xrightarrow{\ell} q)\}_{k\in K} \upharpoonright_r s'!\{l''_k(S''_k).\bar{T''_k}\}_{k\in K}$$

Therefore, it suffices to show that

$$s'!\{l''_k(S''_k).T''_k\}_{k\in K} \sqsubseteq s'!\{l''_k(S''_k).\bar{T''_k}\}_{k\in K}$$

which follows from condition $(G3)$.

- $r = s'$

Follows the same exact proof steps, where instead of sending to s' we receive from s.

- $r \notin \{s, s'\}$

We have

$$s \to s' : \{\ell''_k(S''_k).G''_k\}_{k\in K} \upharpoonright_r \prod_{k\in K} T''_k$$

where $G''_k \upharpoonright_r T''_k$ for $k \in K$. Then, by the inductive hypothesis, we have

$$(G''_k \backslash p \xrightarrow{\ell} q) \upharpoonright_r \bar{T''_k}$$

with $T''_k \sqsubseteq \bar{T''_k}$. Noting that $G' = G\backslash p \xrightarrow{\ell} q = s \to s' : \{\ell''_k(S''_k).(G''_k \backslash p \xrightarrow{\ell} q)\}_{k\in K}$, it follows from Lemma 1.5 that

$$\bar{T} = \prod_{k\in K} \bar{T''_k}$$

is defined with $T \sqsubseteq \bar{T}$. ■